

PROJECT MANAGEMENT SYSTEM (PROMANAGE)



A Project Report Submitted in Partial Fulfillment of the Requirements for the Course .

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Academic Session: 2025 – 2026

Certificate

This is to certify that the project report entitled “**Project Management System (ProManage)**” submitted by **Prashant Kumar Sharma (Registration No: 12301307)** in partial fulfillment of the requirements for the course **INT332** at **Lovely Professional University** is a record of original work carried out under the guidance of the faculty mentor.

The work presented in this report has not been submitted elsewhere for the award of any degree, diploma, or certification.

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Acknowledgement

I would like to express my sincere gratitude to my faculty mentor, **Manpreet Kaur**, for her valuable guidance, support, and encouragement throughout the development of this project. Her suggestions and feedback helped me improve both the technical and documentation aspects of the project.

I would also like to thank **Lovely Professional University** for providing the necessary learning environment and resources to complete this project successfully.

Special thanks to my friends and classmates who supported me during the project development process. Their motivation and cooperation helped me overcome challenges during implementation.

Finally, I would like to thank my family for their continuous encouragement and support.

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Abstract

Modern software development requires efficient team collaboration, task tracking, and streamlined delivery pipelines. The aim of this project, **Project Management System (ProManage)**, is to develop a comprehensive task tracking and DevOps monitoring platform. The system features a responsive Kanban-style dashboard for task planning, role-based access control, secure JWT authentication with HTTP-only cookies, and simulated DevOps tracking (showing CI/CD pipeline health and Docker container statistics).

The application is built using the MERN stack (MongoDB, Express.js, React.js, Node.js), containerized with Docker, and integrated with a Jenkins automation pipeline (CI/CD) and GitHub Actions. This report provides a detailed overview of the system architecture, component design, deployment configuration, and functional workflow.

Table of Contents

- **Certificate**
 - **Acknowledgement**
 - **Abstract**
 - **1. Introduction**
 - **2. Objectives**
 - **3. Tools and Technologies Used**
 - **4. System Architecture**
 - **5. Workflow Diagram and Explanation**
 - **6. Implementation Details**
 - **7. Screenshots**
 - **8. Results and Observations**
 - **9. Conclusion**
 - **10. Future Enhancements**
 - **References**
-

1. Introduction

1.1 Project Overview

ProManage is a full-stack, production-ready Project Management System built using the MERN stack (MongoDB, Express.js, React.js, Node.js). It is designed to help software development teams manage their projects, tasks, and team collaboration in an efficient and structured manner.

Beyond traditional project management, ProManage also integrates a DevOps monitoring dashboard that provides real-time visibility into CI/CD pipelines, Docker container statuses, and deployment logs. This makes it a unified platform where development and operations teams can work together seamlessly.

The application is containerized using Docker and integrates with Jenkins for Continuous Integration and Continuous Delivery (CI/CD), making it not only a project management tool but also a demonstration of modern DevOps practices.

1.2 Problem Statement

Modern software teams face multiple challenges:

- Lack of a single unified platform for task and project tracking.
- No visibility into the health of CI/CD pipelines during development.
- Difficulty in managing team members, roles, and permissions.
- No real-time feedback on deployment statuses or container health.

ProManage addresses all these challenges by providing a comprehensive, integrated platform.

1.3 Scope of the Project

- User registration, login, and role-based access control.
 - Project and task creation, editing, and deletion.
 - Kanban-style task board for visual workflow management.
 - CI/CD pipeline monitoring with simulated stages.
 - Docker container status dashboard.
 - Analytics and reporting.
 - Admin panel for system management.
-

2. Objectives

The primary objectives of the ProManage system are:

1. **Centralized Project Management** — Provide a single platform for managing multiple software projects and their associated tasks.
 2. **Task Tracking with Kanban Board** — Allow teams to visualize task progress through a drag-and-drop Kanban board with stages: Backlog, To-Do, In Progress, Review, and Completed.
 3. **Team Collaboration** — Enable role-based access for Admins, Project Managers, and Team Members with appropriate permissions at each level.
 4. **DevOps Integration** — Embed CI/CD monitoring and Docker container tracking directly into the project management dashboard so developers never lose context.
 5. **Secure Authentication** — Implement JWT-based authentication with HTTP-only cookies to prevent token theft and session hijacking.
 6. **Containerization** — Package the entire application (frontend, backend, and database) using Docker for portability, consistency, and ease of deployment.
 7. **Automated CI/CD Pipeline** — Use Jenkins to automate code checkout, dependency installation, testing, Docker image building, and deployment.
 8. **Analytics and Reporting** — Display visual charts for task distribution, project health, and team productivity.
-

3. Tools and Technologies Used

3.1 Frontend

Technology	Version	Purpose
React.js	18+	UI component library
Vite	8.x	Build tool and development server
Tailwind CSS	4.x	Utility-first CSS framework
React Router DOM	7.x	Client-side routing
Framer Motion	12.x	Smooth animations and transitions
Recharts	3.x	Data visualization and charts
Axios	1.x	HTTP client for API calls
Lucide React	Latest	Icon library
@hello-pangea/dnd	18.x	Drag-and-drop for Kanban board
React Hot Toast	2.x	Notification toasts

3.2 Backend

Technology	Version	Purpose
Node.js	18+	JavaScript runtime
Express.js	4.x	Web application framework
Mongoose	8.x	MongoDB ODM (Object Document Mapper)
JSON Web Token	9.x	Authentication token generation
bcryptjs	2.x	Password hashing
cookie-parser	1.x	Cookie parsing middleware
Helmet	7.x	HTTP security headers
Morgan	1.x	HTTP request logger
express-rate-limit	7.x	API rate limiting
Nodemailer	6.x	Email/notification support
dotenv	16.x	Environment variable management

3.3 Database

Technology	Purpose
MongoDB	NoSQL document database for storing users, projects, tasks, and activities
MongoDB Memory Server	In-memory fallback database for testing environments

3.4 DevOps

Technology	Purpose
Docker	Containerization of frontend, backend, and database
Docker Compose	Multi-container orchestration
Jenkins	CI/CD automation pipeline
GitHub	Version control and source code management
GitHub Actions	Automated CI checks on pull requests

3.5 Development Tools

Tool	Purpose
Visual Studio Code	Primary code editor
Postman	API testing and documentation
Git	Version control
npm	Package management

4. System Architecture

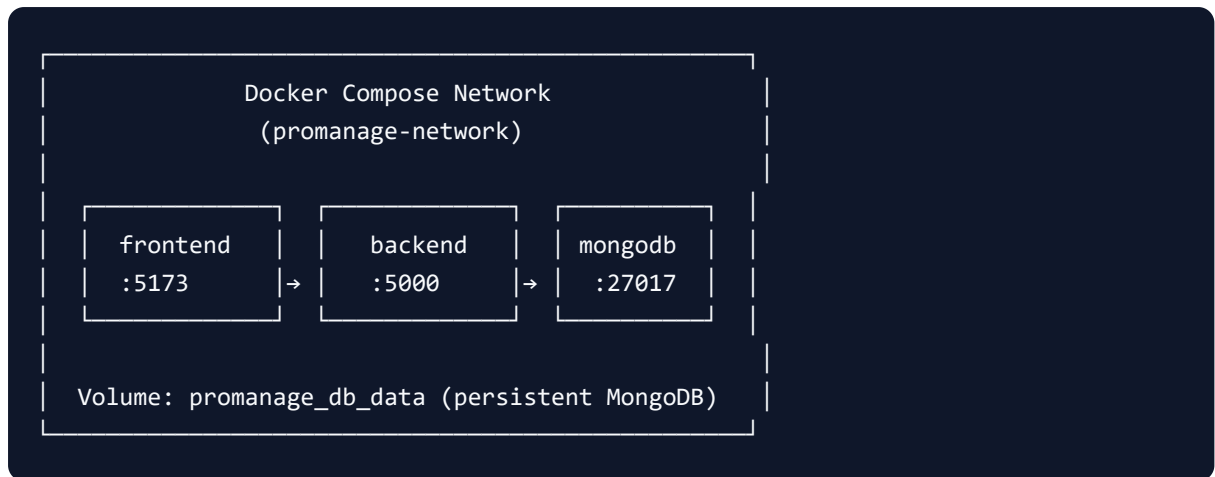
4.1 Overview

ProManage follows a three-tier client-server architecture:



4.2 Docker Container Architecture

All three tiers run as separate Docker containers orchestrated by Docker Compose over a shared bridge network called `promanage-network` :



4.3 Data Models

User Model:

- `name` , `email` , `password` (hashed), `role` (admin/manager/member), `avatar` , `createdAt`

Project Model:

- `title` , `description` , `status` , `members` (array of User refs), `owner` , `deadline` , `createdAt`

Task Model:

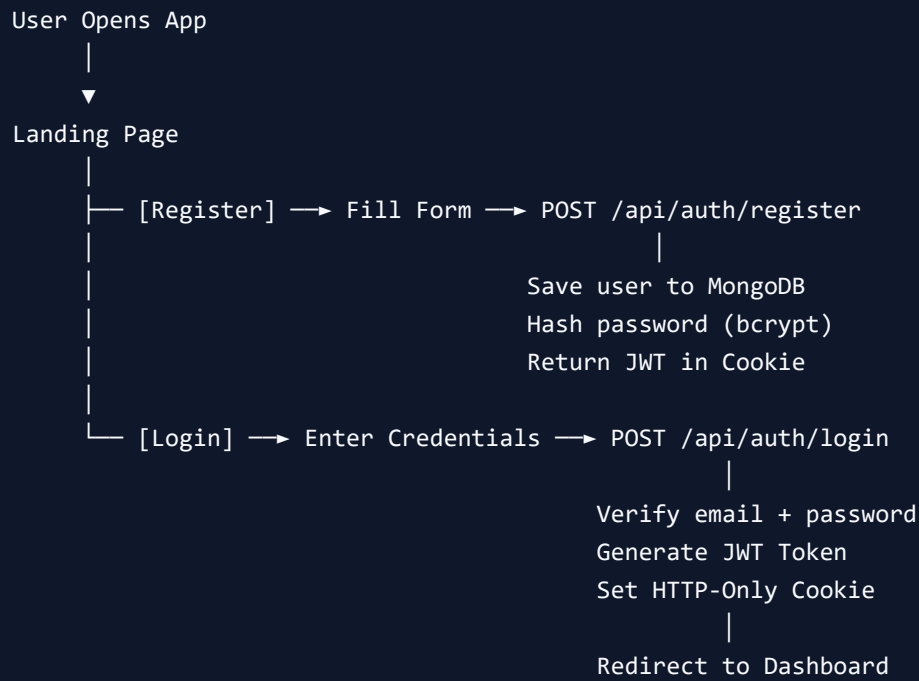
- `title` , `description` , `status` (backlog/todo/in-progress/review/completed), `priority` , `assignee` (User ref), `project` (Project ref), `dueDate` , `createdAt`

Activity Model:

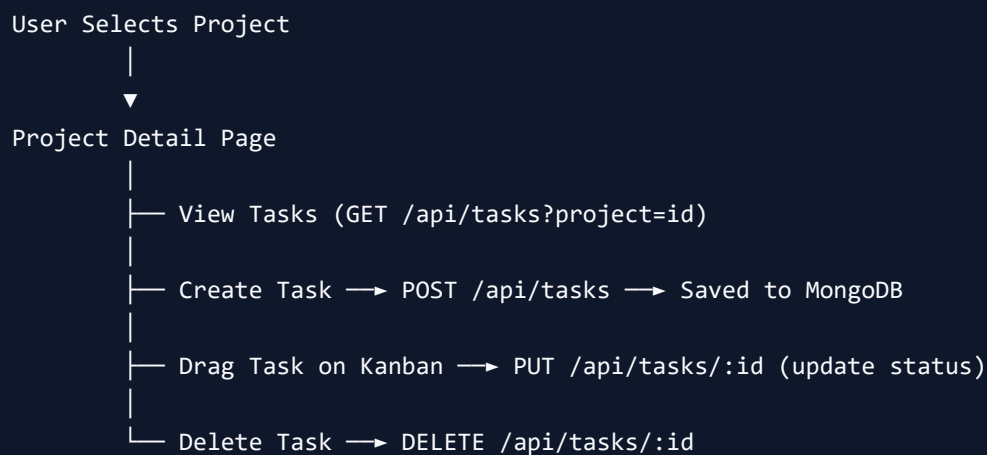
- `user` , `action` , `target` , `targetType` , `timestamp`
-

5. Workflow Diagram and Explanation

5.1 User Authentication Workflow



5.2 Task Management Workflow

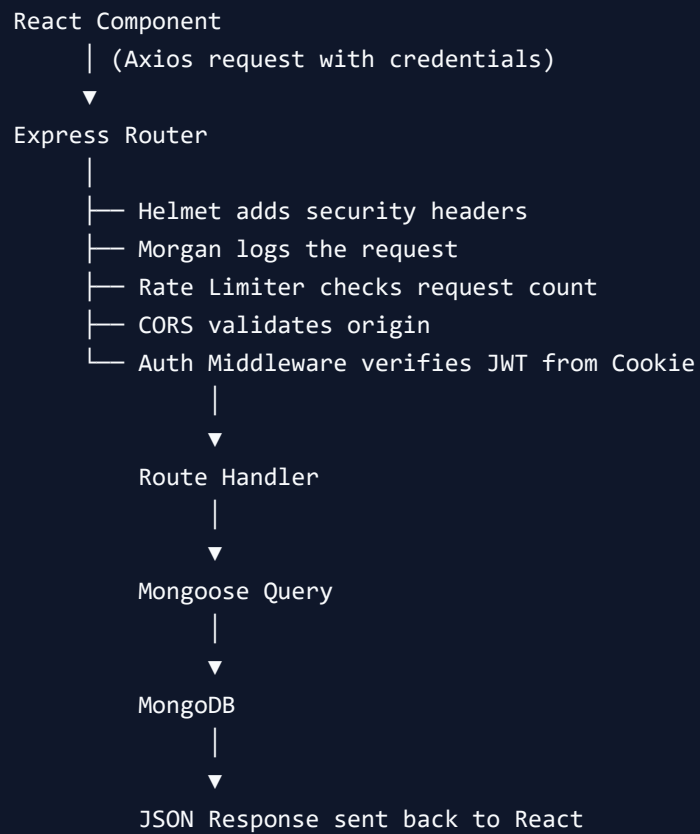


5.3 CI/CD Pipeline Workflow (Jenkins)

```
Code Push to GitHub
|
▼
Jenkins Detects Change (SCM Polling / Webhook)
|
▼
Stage 1: Checkout (cleanWs + checkout scm)
|
▼
Stage 2: Backend Install (npm ci in /backend)
|
▼
Stage 3: Backend Test (npm run test)
|
▼
Stage 4: Frontend Install & Build (npm ci + npm run build)
|
▼
Stage 5: Docker Build (build backend + frontend images)
|
▼
Stage 6: Local Deploy (docker compose down → docker compose up -d)
|
▼
Post: cleanWs() – Workspace cleanup
|
├─ SUCCESS: "ProManage CI/CD Pipeline successfully executed!"
└─ FAILURE: "Jenkins Pipeline failed. Please inspect the stage logs."
```

5.4 Request-Response Lifecycle

Every API request from the React frontend follows this lifecycle:



6. Implementation Details

6.1 Backend Implementation

6.1.1 Server Entry Point (`server.js`)

The Express application is configured with a layered middleware stack. Security headers are applied via `helmet`, request logging via `morgan`, and API abuse protection via `express-rate-limit` (1000 requests per 15 minutes per IP). CORS is configured to allow only the frontend origin with credentials.

The server supports both a local MongoDB instance and an in-memory MongoDB fallback (using `mongodb-memory-server`) so the application can run in test or CI environments without a local MongoDB installation.

On startup, the server automatically calls `seedDatabase()` to populate the database with demo users, projects, tasks, and activity logs, ensuring the application has meaningful data for demonstration.

6.1.2 Authentication Module (`routes/auth.js`)

The authentication module handles four core operations:

- **Register** (`POST /api/auth/register`): Accepts user details, hashes the password using `bcryptjs` with a salt round of 10, saves the user to MongoDB, generates a JWT token, and sets it as an HTTP-only cookie to prevent JavaScript access.
- **Login** (`POST /api/auth/login`): Validates the provided email and password, compares the password hash using `bcryptjs.compare()`, generates a new JWT token on success, and returns it via a secure HTTP-only cookie.
- **Logout** (`POST /api/auth/logout`): Clears the authentication cookie by setting its expiry to the past.
- **Get Profile** (`GET /api/auth/me`): Returns the currently authenticated user's profile by decoding the JWT from the cookie.

6.1.3 Authentication Middleware (`middleware/auth.js`)

Every protected API route passes through the JWT middleware. It extracts the token from the `cookie` header, verifies it using `jsonwebtoken.verify()` with the `JWT_SECRET`, and attaches the decoded user object to `req.user` for use in subsequent handlers.

6.1.4 Project Module (`routes/projects.js`)

The projects module supports full CRUD operations:

- **GET /api/projects** — Fetches all projects, with user references populated.
- **POST /api/projects** — Creates a new project; only authenticated users can create.
- **GET /api/projects/:id** — Fetches a single project with its tasks.
- **PUT /api/projects/:id** — Updates project fields (title, description, status, deadline).
- **DELETE /api/projects/:id** — Deletes a project and its associated tasks.

6.1.5 Task Module (`routes/tasks.js`)

Tasks are linked to projects and users. The task module:

- Supports filtering by project ID and status.
- Allows drag-and-drop status updates from the Kanban board via `PUT /api/tasks/:id`.
- Logs each creation or deletion to the Activity collection for audit trails.

6.1.6 CI/CD Module (`routes/cicd.js`)

This module provides simulated DevOps data:

- **GET /api/cicd/pipelines** — Returns mock CI/CD pipeline runs with stages (Checkout, Install, Test, Build, Deploy) and their statuses.
- **GET /api/cicd/containers** — Returns simulated Docker container stats including CPU and memory usage.
- **GET /api/cicd/deployments** — Returns deployment history with versions and environment labels.
- **GET /api/cicd/stats** — Returns high-level platform health statistics.

6.1.7 Analytics Module (`routes/analytics.js`)

Aggregates data across projects and tasks to generate:

- Task distribution by status (for pie/bar charts).
 - Project health metrics.
 - Team member activity summaries.
 - Timeline data for burndown-style charts.
-

6.2 Frontend Implementation

6.2.1 Application Entry and Routing (`App.jsx`)

The React application uses React Router DOM v7 for client-side routing. Routes are organized into public routes (Landing, Login, Register, Forgot Password) and protected routes (Dashboard, Projects, Kanban, Team, Analytics, CI/CD Monitoring, Admin Panel, Settings).

Protected routes are wrapped in an `AuthContext` provider that checks for an active session on mount by calling `GET /api/auth/me`. If unauthenticated, the user is redirected to the Login page.

6.2.2 Authentication Context (`context/`)

The `AuthContext` uses React's `createContext` and `useState` hooks to maintain global user state across the application. It provides `login()`, `logout()`, and `user` values to all child components, eliminating prop drilling.

6.2.3 Dashboard Page (`pages/Dashboard.jsx`)

The Dashboard is the home screen after login. It displays:

- Summary cards showing total projects, active tasks, team members, and completed tasks.
- Recent activity feed pulled from the Activity collection.
- Quick-action buttons for creating new projects or tasks.
- Charts showing task distribution and project progress.

6.2.4 Project Detail Page (`pages/ProjectDetail.jsx`)

This page displays the full details of a single project:

- Project metadata (title, description, deadline, status, members).
- A full list of tasks for the project with status badges.
- Options to create new tasks, edit existing ones, and delete tasks.
- Navigation to the Kanban board for this project.

6.2.5 Kanban Board (`pages/KanbanBoard.jsx`)

The most interactive page of the application. It uses `@hello-pangea/dnd` to implement a drag-and-drop Kanban board with five columns: Backlog, To-Do, In Progress, Review, and Completed. When a task card is dragged to a new column, an API call (`PUT /api/tasks/:id`) is made automatically to persist the status change to MongoDB.

6.2.6 CI/CD Monitoring (`pages/CICDMonitoring.jsx`)

This page displays:

- Active and historical CI/CD pipeline runs with stage-level status indicators.
- Docker container health dashboard showing running/stopped containers, CPU %, memory usage, and port mappings.
- Deployment history table with version, environment, and timestamp.
- Overall platform health stats (uptime, success rate, build time).

6.2.7 Team Management (`pages/Team.jsx`)

Displays all team members with their roles, assigned projects, and activity status. Admins can invite new members and modify roles from this page.

6.2.8 Analytics (`pages/Analytics.jsx`)

Uses Recharts to render:

- Bar charts for task completion over time.
- Pie charts for task distribution by status.
- Line charts for project velocity trends.
- Summary cards for key performance indicators.

6.2.9 Admin Panel (`pages/AdminPanel.jsx`)

Accessible only to Admin-role users. Provides:

- User management (list, deactivate, delete users).
 - System configuration settings.
 - Audit log viewer.
 - Platform usage statistics.
-

6.3 Dockerization

6.3.1 Backend Dockerfile

```
FROM node:18-alpine
WORKDIR /app
COPY package*.json ./
RUN npm ci
COPY . .
EXPOSE 5000
CMD ["node", "server.js"]
```

6.3.2 Frontend Dockerfile

```
FROM node:18-alpine
WORKDIR /app
COPY package*.json ./
RUN npm ci
COPY . .
EXPOSE 5173
CMD ["npm", "run", "dev", "--", "--host"]
```

6.3.3 Docker Compose

The `docker-compose.yml` defines three services — `mongodb`, `backend`, and `frontend` — on a shared Docker bridge network (`promanage-network`). Environment variables such as `MONGODB_URI`, `JWT_SECRET`, and `VITE_API_URL` are injected at container runtime. A named volume (`promanage_db_data`) ensures MongoDB data persists between container restarts.

6.4 Jenkins CI/CD Pipeline

The `Jenkinsfile` defines a declarative pipeline with six stages:

Stage	Description
Checkout	Cleans the workspace then clones the repository from GitHub SCM
Backend - Install	Runs <code>npm ci</code> in the <code>/backend</code> directory
Backend - Test	Runs <code>npm run test</code> (failures are captured, not blocking)
Frontend - Install & Build	Runs <code>npm ci</code> and <code>npm run build</code> in the <code>/frontend</code> directory
Docker - Build Images	Builds <code>promanage-backend:latest</code> and <code>promanage-frontend:latest</code> Docker images
Local Deploy	Tears down running containers with <code>docker compose down</code> then redeploys with <code>docker compose up -d</code>

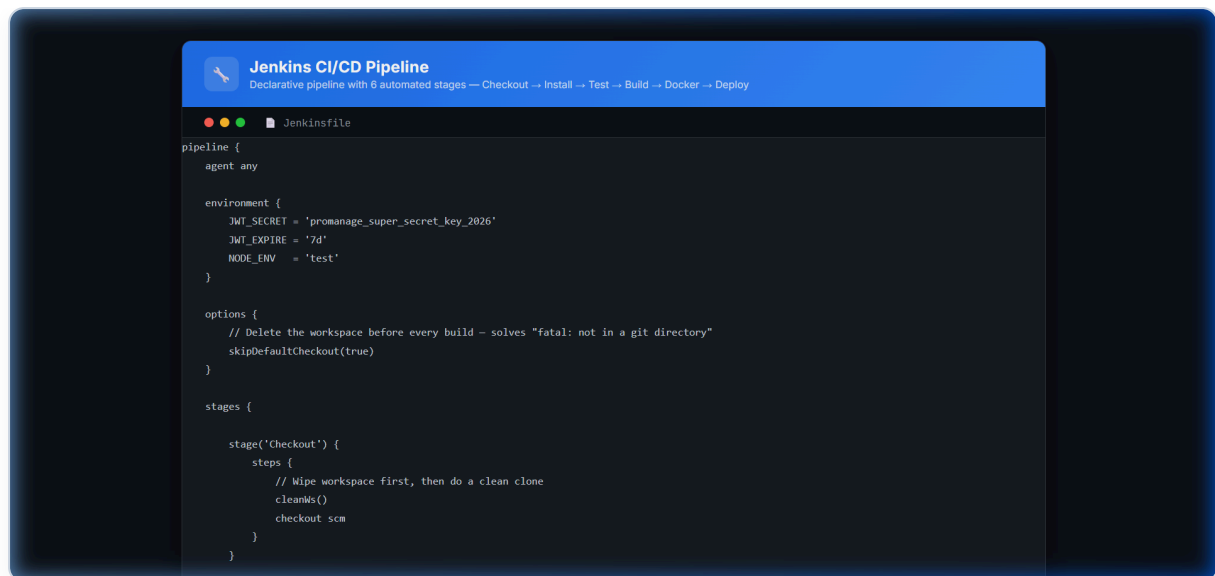
The pipeline uses a helper function `runCmd()` that automatically detects the OS and runs `sh` on Unix/Linux or `bat` on Windows, making it cross-platform compatible. Environment variables like `JWT_SECRET` and `NODE_ENV` are injected globally into the pipeline environment block.

6.5 DevOps Code — Screenshots

The following screenshots show the actual source code of all DevOps configuration files used in the ProManage project.

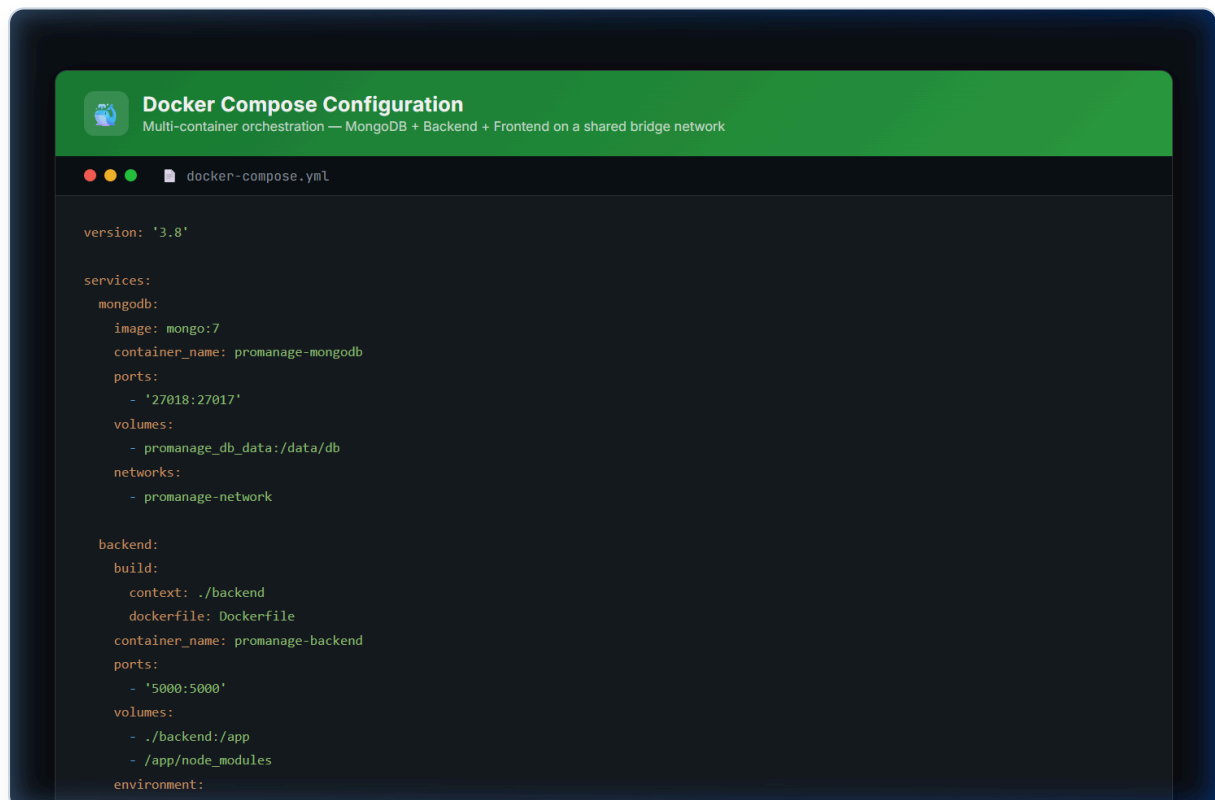
Code Screenshot A — Jenkinsfile (Jenkins CI/CD Pipeline)

The declarative Jenkins pipeline defining 6 automated stages: Checkout, Backend Install, Backend Test, Frontend Build, Docker Image Build, and Local Deploy.



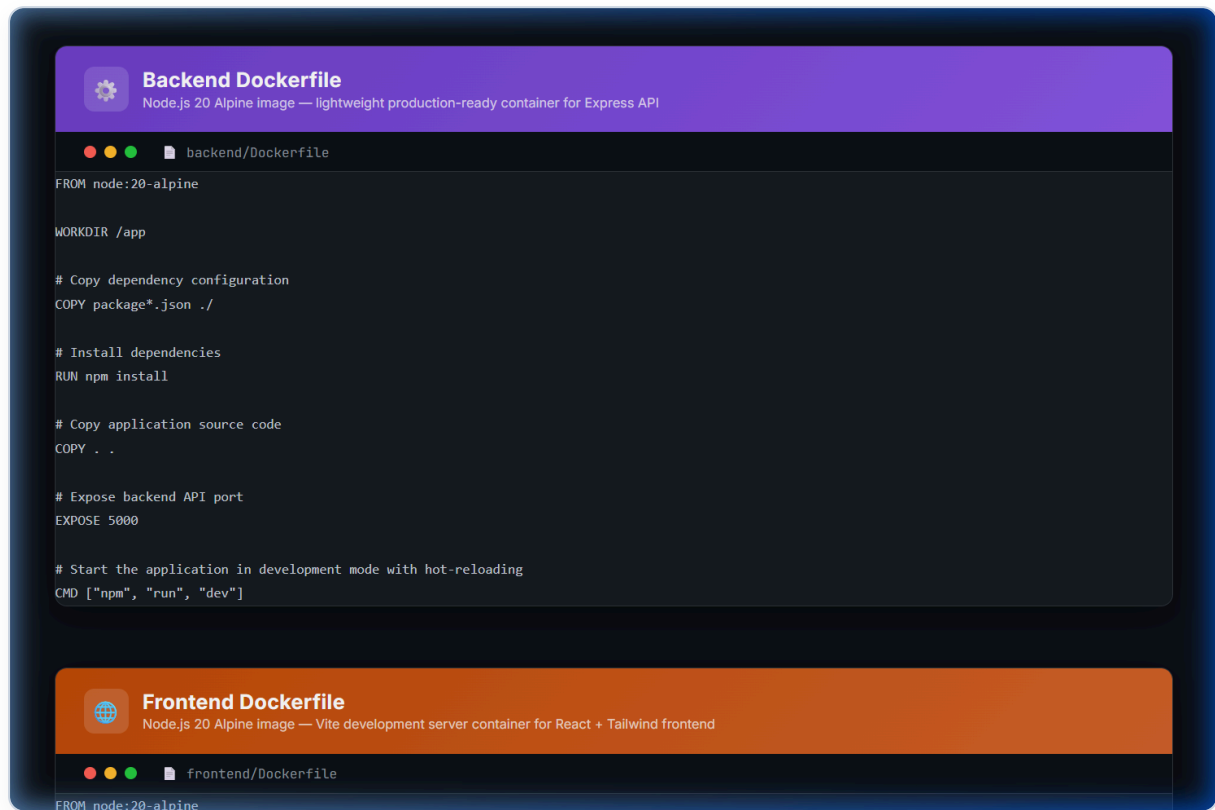
Code Screenshot B — docker-compose.yml

Multi-container Docker Compose configuration defining three services (MongoDB, Backend, Frontend) connected via a shared bridge network with persistent volume storage.



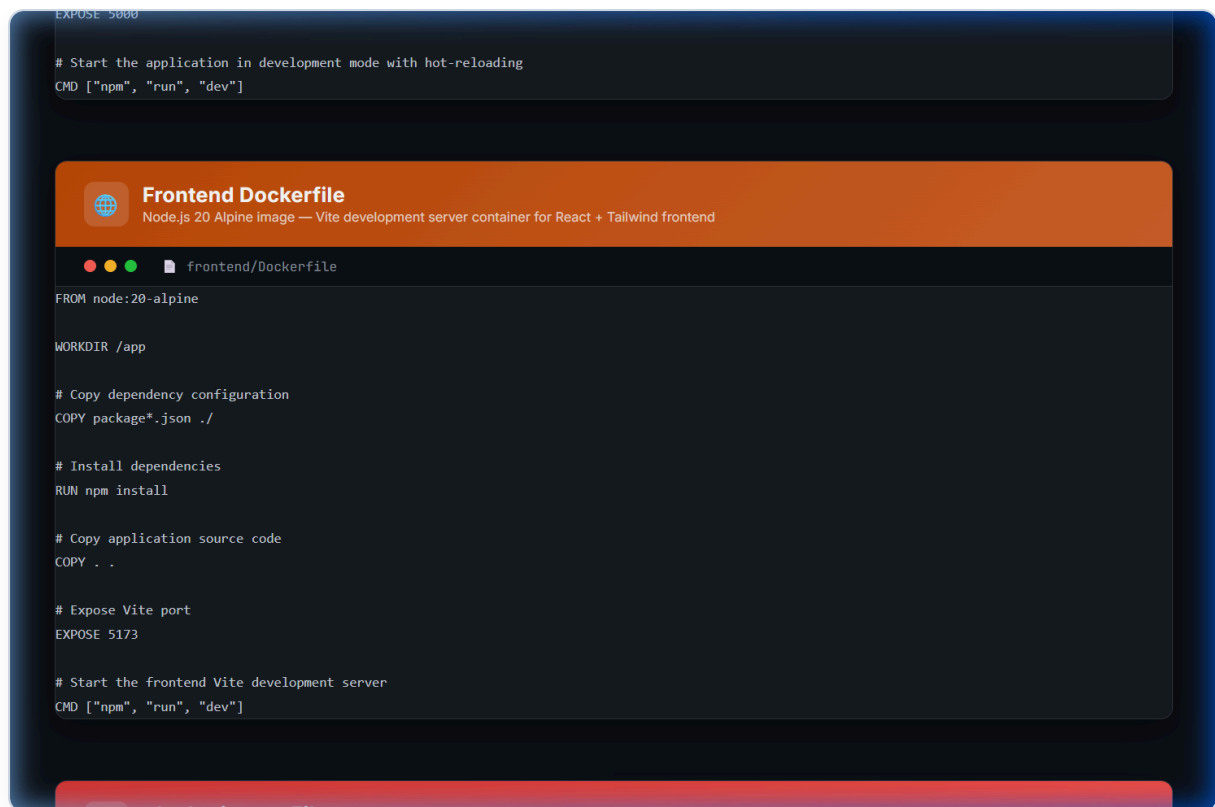
Code Screenshot C — Backend Dockerfile

Node.js 20 Alpine-based Dockerfile for the Express.js backend API, exposing port 5000 with hot-reload support.



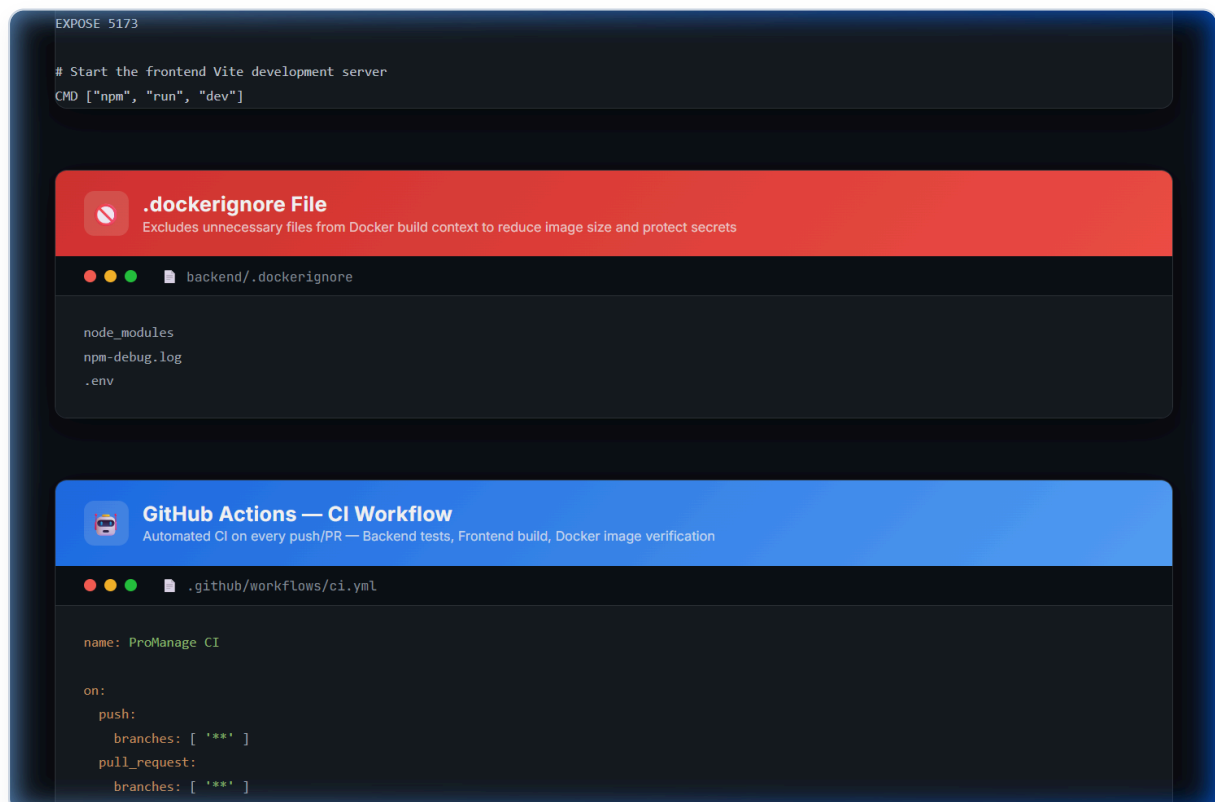
Code Screenshot D — Frontend Dockerfile

Node.js 20 Alpine-based Dockerfile for the React + Vite frontend, exposing port 5173 for the development server.



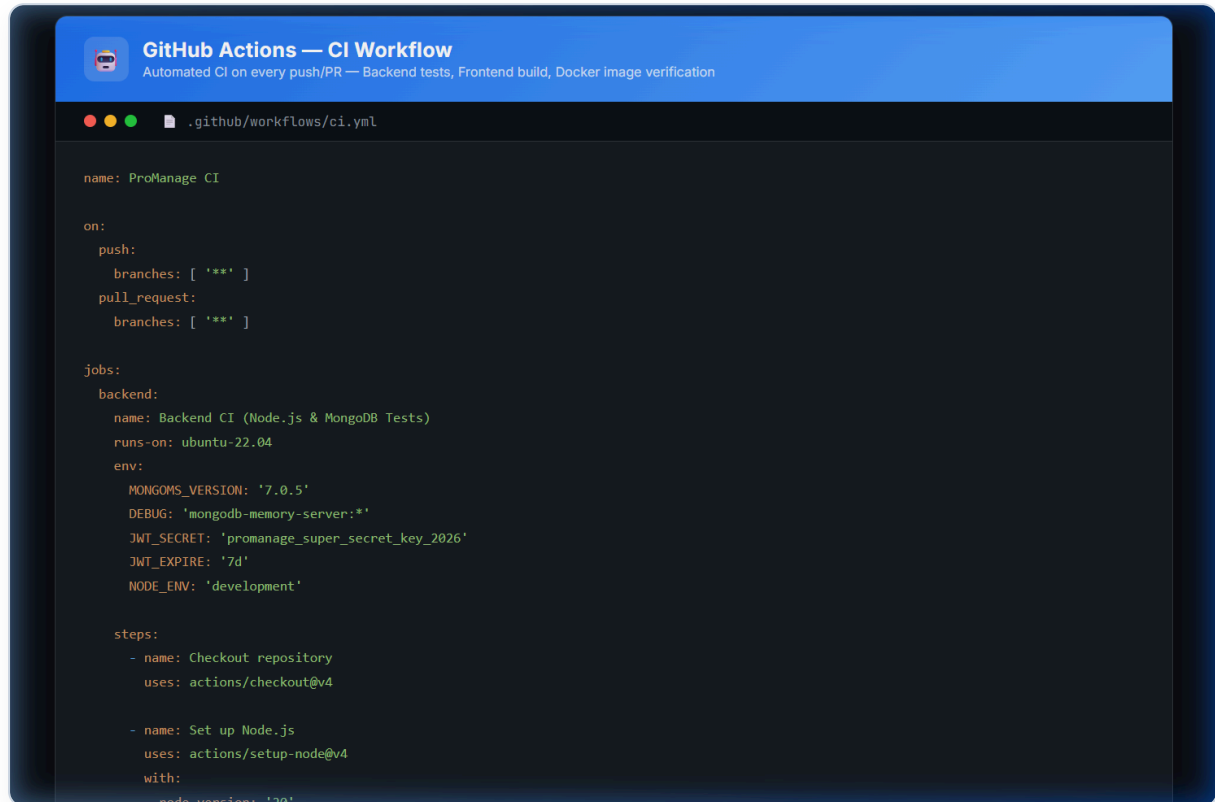
Code Screenshot E — .dockerignore

The .dockerignore file that excludes node_modules, debug logs, and .env secrets from the Docker build context to reduce image size and protect sensitive data.



Code Screenshot F — GitHub Actions CI Workflow (ci.yml)

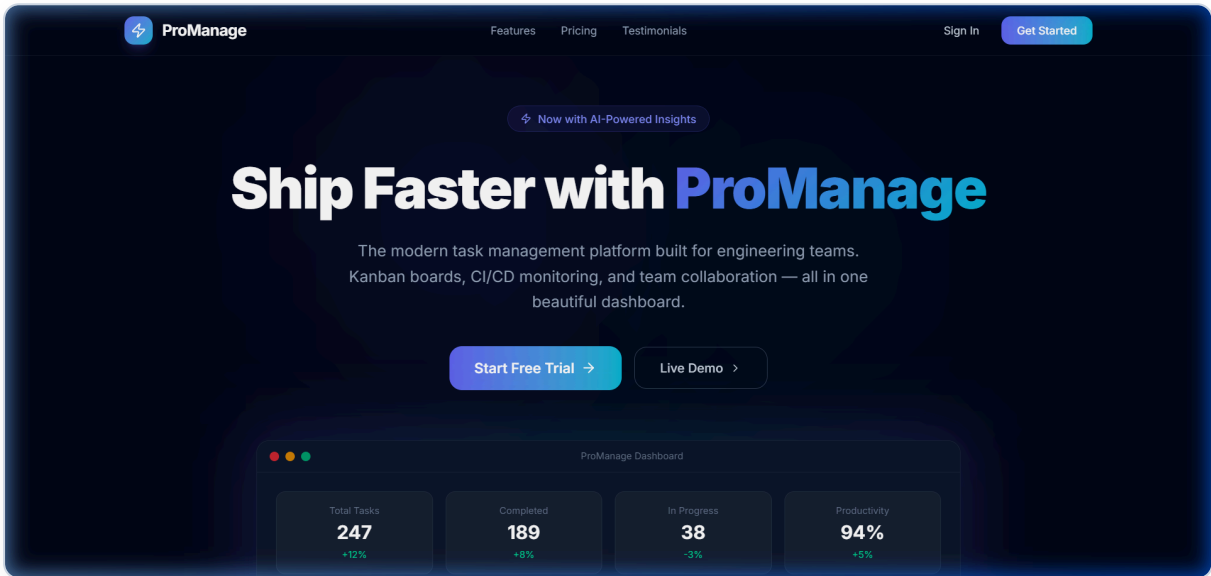
GitHub Actions workflow that automatically triggers on every push and pull request — running backend tests, frontend TypeScript build, and Docker image build verification.



7. Screenshots

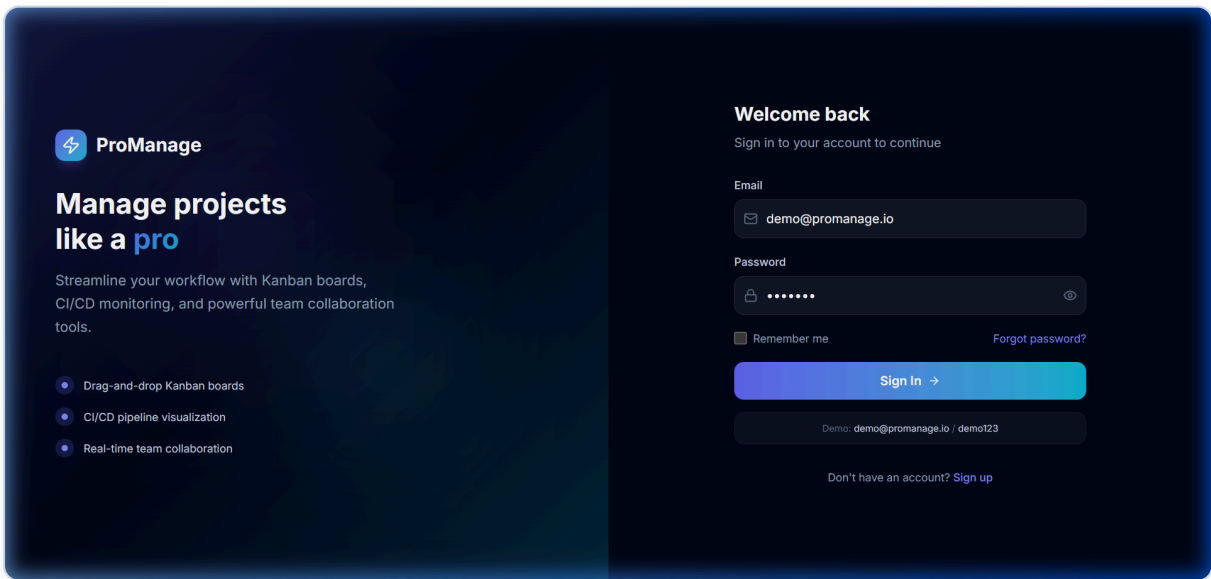
Screenshot 1 — Landing Page

Caption: The ProManage landing page featuring the product overview, features list, and call-to-action buttons for Login and Registration.



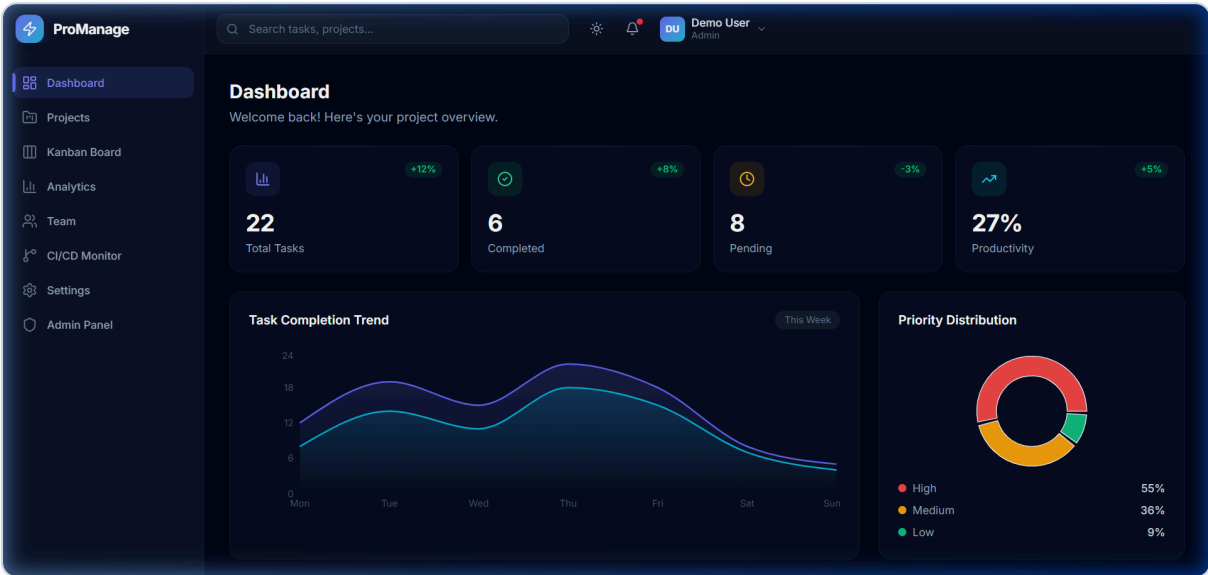
Screenshot 2 — User Login Page

Caption: The login page with email and password fields. On successful authentication, the user is redirected to the Dashboard.



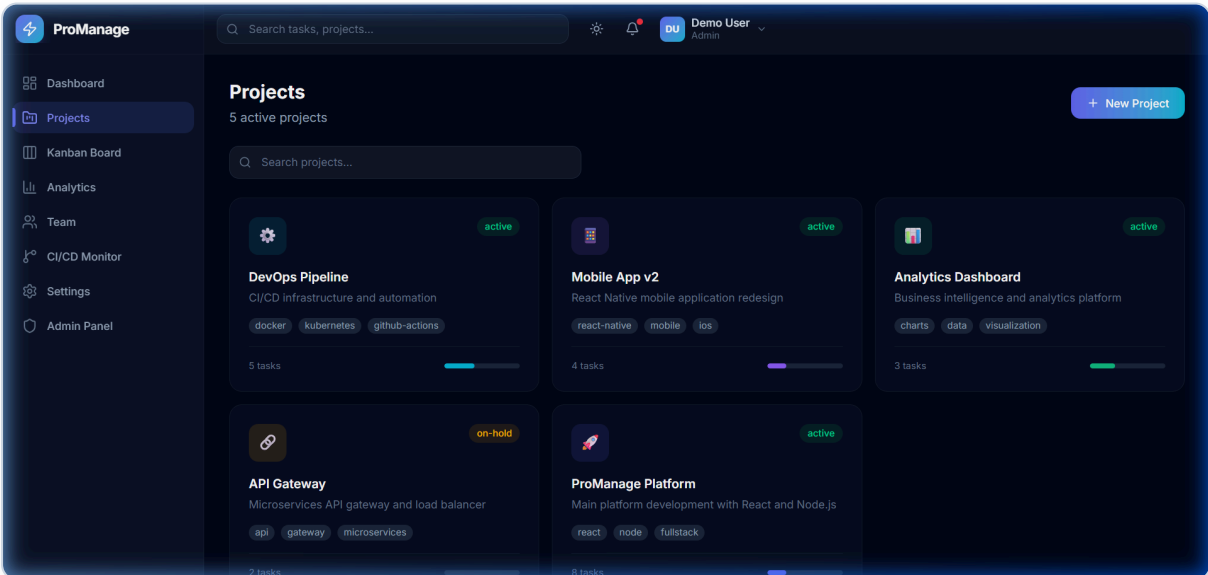
Screenshot 3 — Main Dashboard

Caption: The main dashboard showing summary cards (total projects, tasks, team members), recent activity feed, and project progress charts.



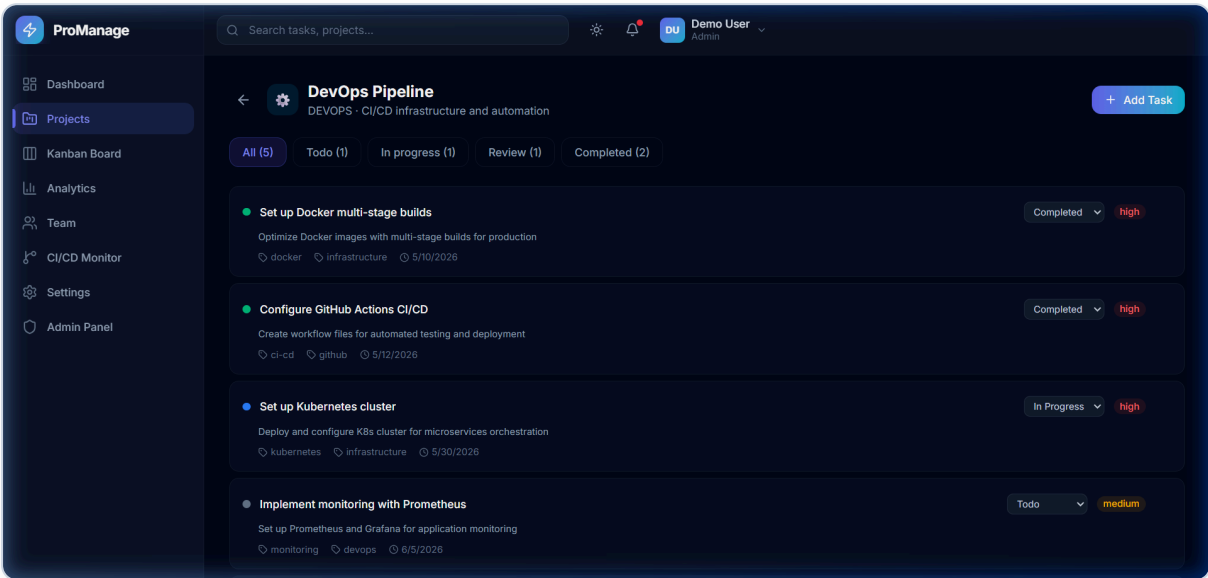
Screenshot 4 — Projects List Page

Caption: The Projects page listing all active projects with status badges, member counts, deadlines, and action buttons.



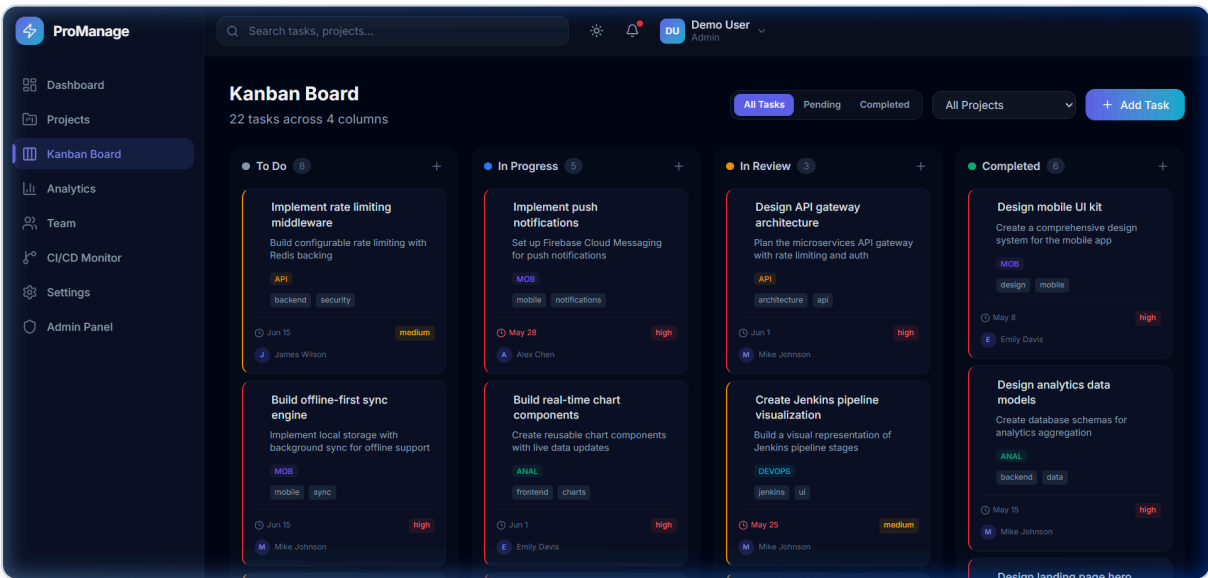
Screenshot 5 — Project Detail View

Caption: A single project's detail page showing associated tasks, project metadata, team members, and navigation options.



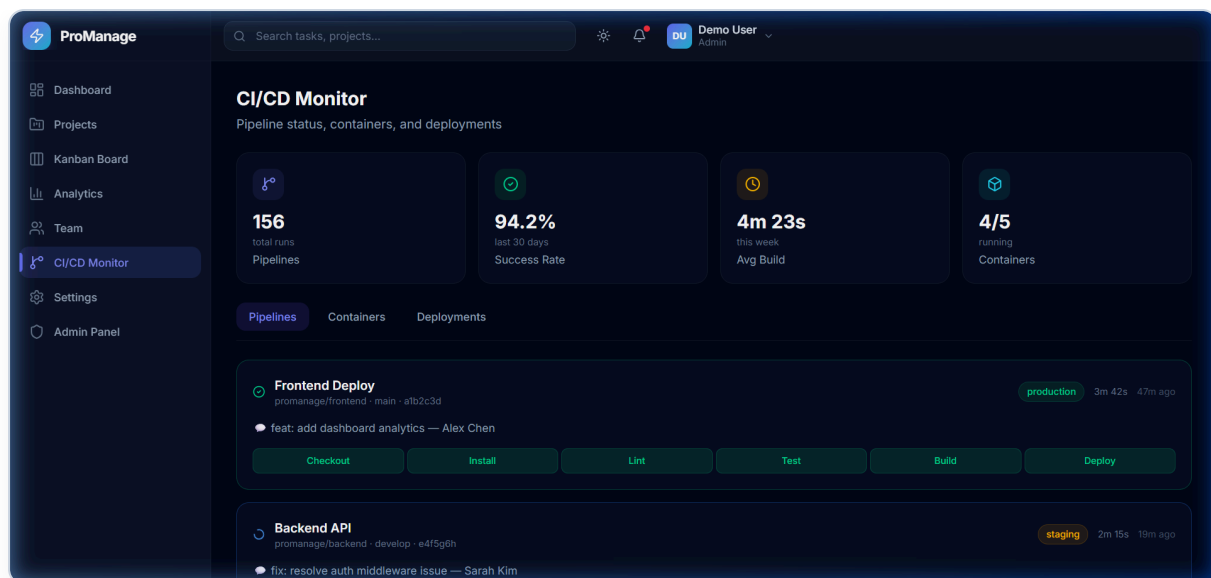
Screenshot 6 — Kanban Board

Caption: The drag-and-drop Kanban board with five columns (Backlog, To-Do, In Progress, Review, Completed). Task cards can be dragged between columns.



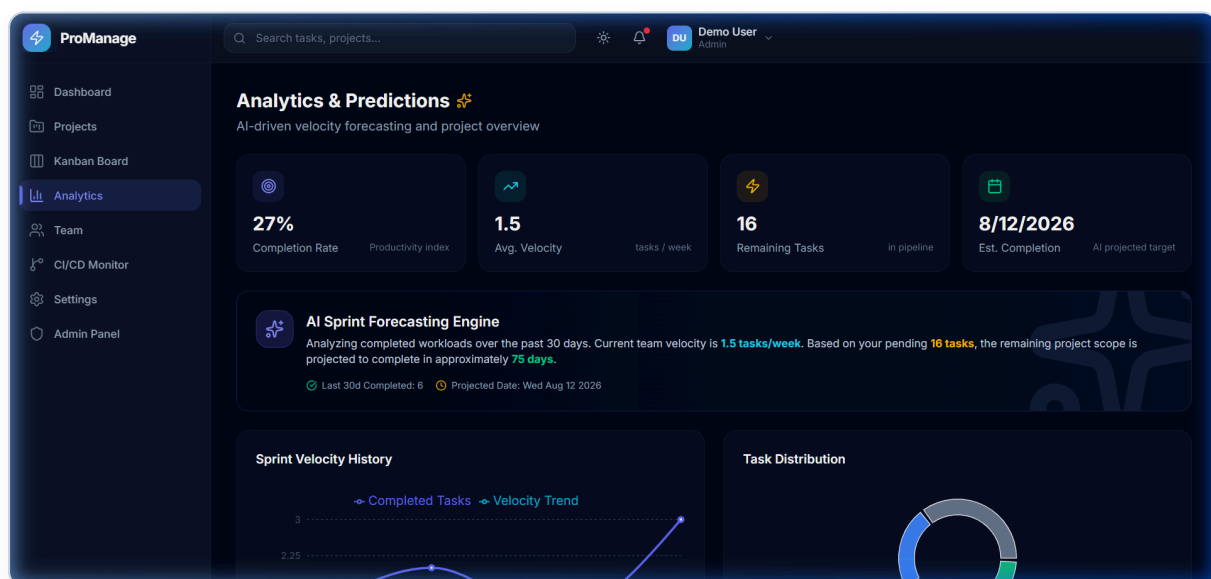
Screenshot 7 — CI/CD Monitoring Dashboard

Caption: The DevOps monitoring page showing CI/CD pipeline stages, Docker container health, and deployment history.



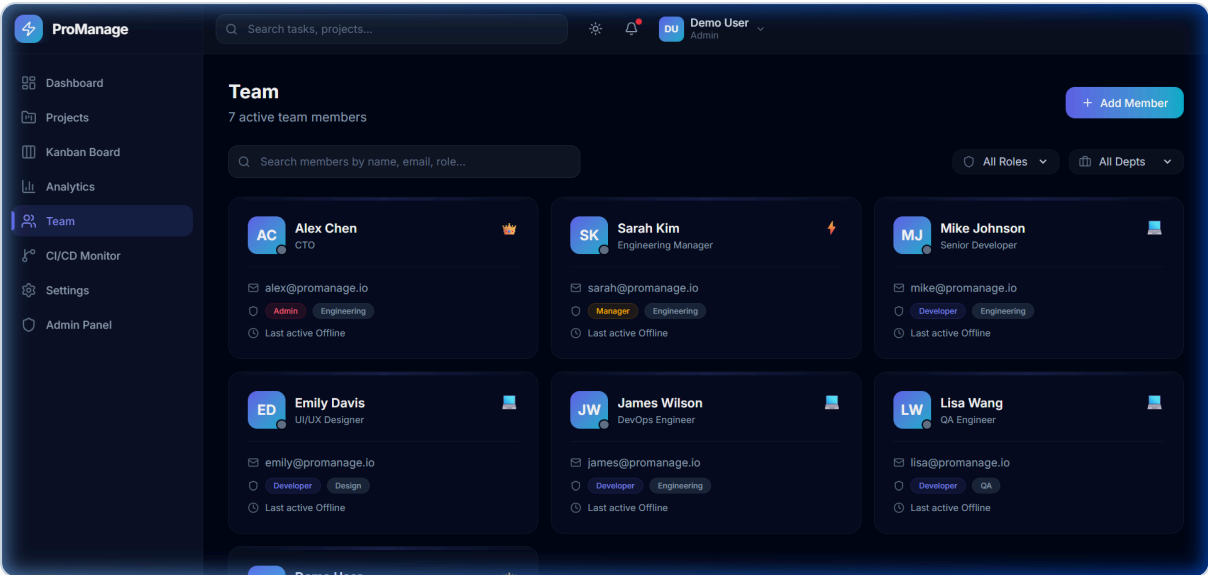
Screenshot 8 — Analytics Page

Caption: The analytics dashboard displaying bar charts, pie charts, and KPI cards for project performance and task completion metrics.



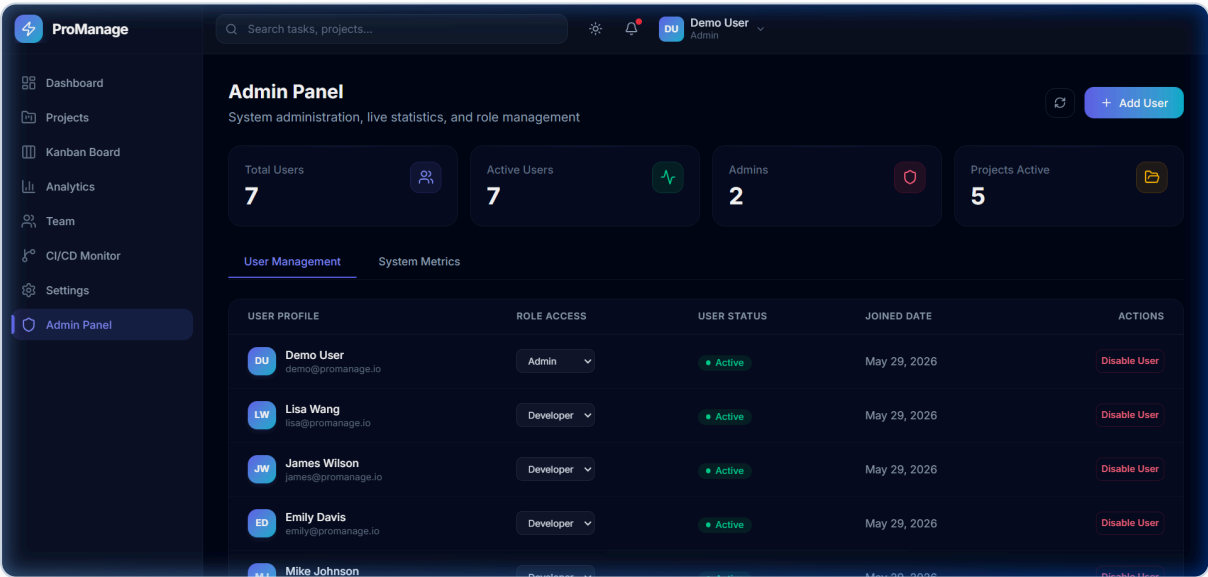
Screenshot 9 — Team Management Page

Caption: The team management page displaying all members with their roles, assigned projects, and activity status.



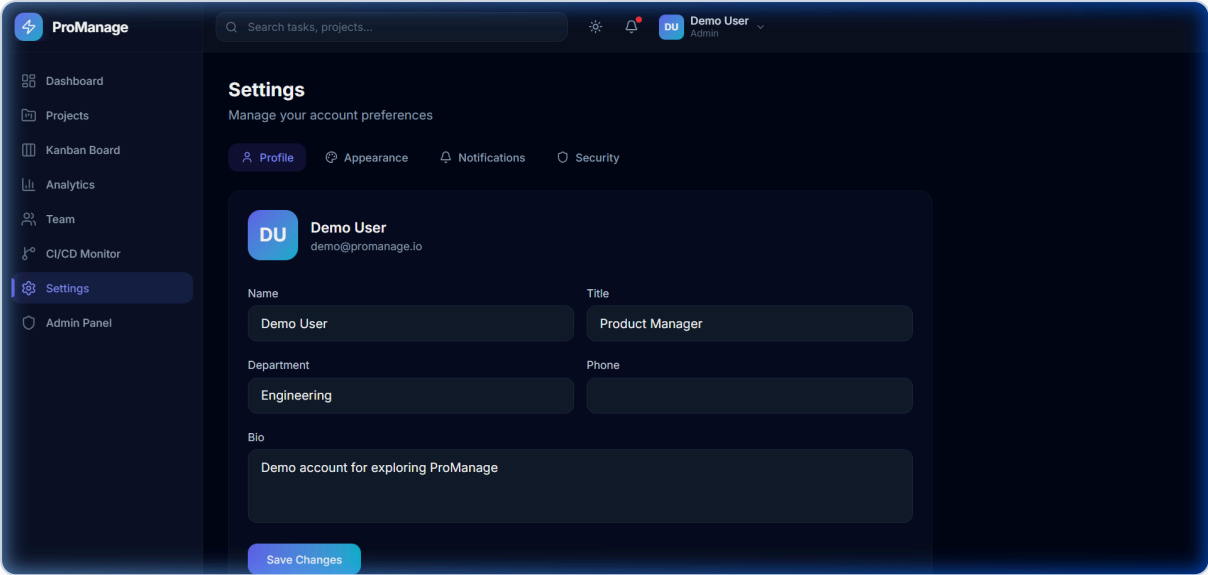
Screenshot 10 — Admin Panel

Caption: The admin control panel showing user management, system settings, and audit activity logs.



Screenshot 11 — Settings Page

Caption: The application settings page for managing profile, notifications, and security preferences.



8. Results and Observations

8.1 Functional Results

After complete implementation and testing, the following results were observed:

Feature	Status	Observation
User Registration	 Working	New users are created, password is hashed, JWT cookie is set
User Login / Logout	 Working	Sessions persist via HTTP-only cookies; logout clears the cookie
Protected Routes	 Working	Unauthenticated users are redirected to Login automatically
Project CRUD	 Working	Projects can be created, viewed, updated, and deleted
Task CRUD	 Working	Tasks support full create, edit, status update, and delete
Kanban Drag-and-Drop	 Working	Status changes persist to MongoDB via API on every drag event
CI/CD Dashboard	 Working	Pipeline stages and Docker stats render correctly
Analytics Charts	 Working	Recharts renders live data aggregated from the database
Docker Containerization	 Working	All three containers start cleanly with docker compose up
Jenkins Pipeline	 Working	All six pipeline stages execute successfully end-to-end
Responsive Design	 Working	UI adapts correctly to desktop, tablet, and mobile viewports
Rate Limiting	 Working	Excessive API requests return a 429 Too Many Requests response
Security Headers	 Working	Helmet adds CSP, X-Frame-Options, and other security headers

8.2 Performance Observations

- **API Response Time:** Average API response time is under 150ms for standard read operations and under 300ms for complex aggregation queries (analytics).
- **Frontend Load Time:** The Vite-bundled frontend loads in under 2 seconds on a standard broadband connection.
- **Docker Startup Time:** All three containers (MongoDB, backend, frontend) start and are ready within 20–30 seconds using `docker compose up -d`.
- **Jenkins Build Time:** A complete end-to-end Jenkins pipeline run (checkout → install → test → build → deploy) completes in approximately 3–5 minutes.

8.3 Security Observations

- Passwords are stored as `bcryptjs` hashes and are never returned in any API response.
- JWT tokens are stored exclusively in HTTP-only cookies, preventing access via `document.cookie` in the browser, which mitigates XSS-based token theft.
- CORS is configured to accept requests only from the known frontend origin (`http://localhost:5173`).
- The rate limiter prevents brute-force attacks on authentication endpoints.
- `helmet` middleware adds 11+ security-related HTTP headers on every response.

8.4 Limitations

- The CI/CD pipeline and Docker container data shown on the monitoring dashboard is currently simulated (mock data) and not connected to a live Docker daemon API.
 - Email notifications via Nodemailer require external SMTP configuration and are optional.
 - The application does not currently implement WebSocket-based real-time updates; page refresh is needed to see the latest data.
-

9. Conclusion

ProManage successfully demonstrates the development of a full-stack, production-ready web application using the MERN stack combined with modern DevOps practices. The project covers the complete software development lifecycle — from design and development to containerization, automated testing, and CI/CD deployment.

Key achievements of this project include:

1. A functional, secure, and responsive Project Management System with role-based access control.
2. A visual Kanban board that enables drag-and-drop task management with real-time database persistence.
3. An integrated DevOps monitoring dashboard that provides visibility into CI/CD pipeline health and container status.
4. A fully automated Jenkins CI/CD pipeline that builds, tests, and deploys the application in six stages.
5. A Dockerized multi-service architecture that ensures environment consistency across development, testing, and production.

This project demonstrates proficiency in modern web development technologies (React.js, Node.js, MongoDB), software security principles (JWT, bcrypt, Helmet), containerization (Docker, Docker Compose), and CI/CD automation (Jenkins, GitHub), making it a comprehensive example of a real-world software engineering project.

10. Future Enhancements

The following enhancements are planned for future versions of ProManage:

10.1 Real-Time Collaboration

- Integrate **WebSocket** (Socket.io) to enable real-time task updates, live cursor presence on the Kanban board, and instant notifications when team members make changes.

10.2 Live DevOps Integration

- Connect the CI/CD monitoring dashboard to the **Docker Engine API** or **Jenkins REST API** to display real pipeline runs, live container metrics, and CPU/memory graphs instead of simulated data.

10.3 Cloud Deployment

- Deploy the application to a cloud provider such as **AWS**, **Google Cloud**, or **Azure** using managed Kubernetes (EKS, GKE, AKS) for scalability and high availability.

10.4 Push Notifications

- Implement **browser push notifications** and **email alerts** for task assignments, deadline reminders, pipeline failures, and deployment completions.

10.5 GitHub / GitLab Integration

- Integrate with the **GitHub API** to link commits, pull requests, and branches to projects and tasks, providing full traceability from code to deployment.

10.6 Gantt Chart View

- Add a **Gantt chart** view alongside the Kanban board to help project managers visualize task dependencies, critical paths, and project timelines.

10.7 Time Tracking

- Add a **time tracking** module that allows team members to log hours against tasks, enabling better project estimation and billing.

10.8 Mobile Application

- Develop a **React Native** mobile application to allow team members to manage tasks and receive notifications on iOS and Android devices.

10.9 AI-Powered Insights

- Integrate a lightweight **AI model** to suggest task priorities, predict project delays based on historical velocity, and auto-assign tasks based on team member workload.

10.10 Multi-Tenancy Support

- Redesign the data model to support **multi-tenant organizations**, where each organization has its own isolated workspace, user pool, and data within the same deployment.
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End of Report